

Instructions

This exam is **20 pages** long, and will last **120 minutes**. Before the exam, please read this entire page, then answer the question at the bottom. Do **not** open the test until instructed!

- All outside resources are prohibited, including but not limited to, books, other people, or anything with internet access. Sole exception is one hand-written (not iPad or printed), double-sided, letter-sized cheat sheet.
- The problems are listed in no particular order of difficulty. Make sure to read all of the problems before getting started.
- There are blank pages at the end of the exam for scratch work. If you would like work on that page to be graded, please clearly indicate so on the problem. **Do not tear them off.**
- Do not do more than what is asked of you. For example, if we do not ask for a proof of correctness for your algorithm, you do not need to provide one. Indecipherable answers will receive no credit.
- Any algorithm covered in lecture may be used as a black box, which means you can include it in your algorithms and talk about its correctness/runtime without proof.
- After the exam, please **do not (publicly) discuss the exam material** until solutions are released. There are some students taking the exam on a later date!

Good luck! You got this.

(a) Please write your name below.

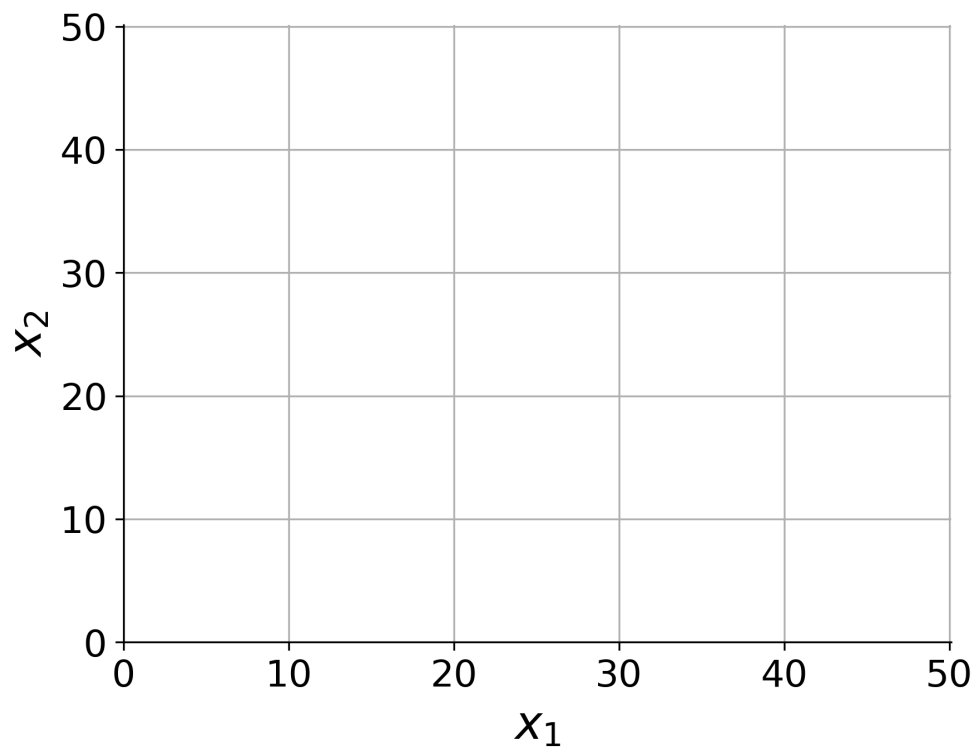
(b) Please write your ChicagoID (e.g. 12345678A) below.

PROBLEM 1 (Linear Programming (12 points)). A factory produces two types of laptops, named x_1 and x_2 . When sold, they make a net profit of \$600 for each unit of x_1 and \$400 for each unit of x_2 . Laptops are produced using two machines, A and B . Producing one unit of x_1 requires 5 minutes on machine A AND 3 minutes on machine B . Producing one unit of x_2 requires 6 minutes on machine B . Due to labor union rules, the machine operators can only run machine A for 100 minutes per day and machine B for 120 minutes per day. Since laptops are popular, assume that the factory can sell all units of x_1 and x_2 produced. You may also assume that the factory can produce and sell fractional amounts of x_1 and x_2 .

- (a) (4 points) Formulate this problem as a linear program in two variables that finds the maximum profit that can be obtained per day.

- (b) (4 points) Formulate the dual problem.

- (c) (4 points) Solve your LP geometrically: plot the feasible region, give the coordinates of every vertex, and circle the vertex maximizing profit. What is the maximum profit possible?



PROBLEM 2 (Fire engine routing (16 points)). You are given a directed, weighted graph $G = (V, E, w)$ of the roads of Hyde Park. Each node is an intersection and each edge is a road connecting two intersections. The time it takes to travel a road e is the weight w_e . Note that some roads are **one ways**. Chicago Fire Department Engine Company 60 is located at 55th and University Ave, which is node **S** in G . Fire Engines are allowed to go the wrong way (for example driving south on University Ave) in an emergency. However, traveling a one way edge in the **wrong direction** incurs an additional cost of p .

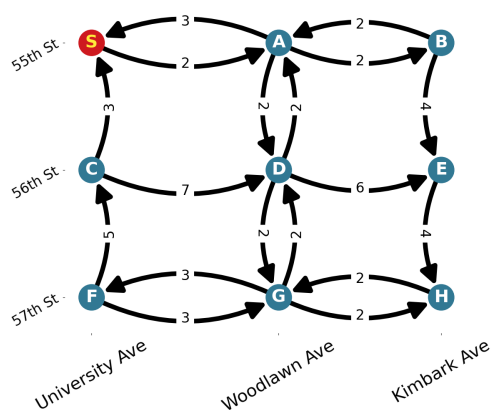


Figure 1: One way streets in Hyde Park

For example, if $p = 10$ it is cheaper to go south on University to reach 56th & University for a cost of $3 + 10 = 13$ instead of looping around through Woodlawn and up from 57th for a total cost of $2 + 2 + 2 + 3 + 5 = 14$.

Create a graph $G_p = (V_p, E_p, w_p)$ such that calling Dijkstra's algorithm on G_p will give the minimum distance paths from s to all other vertices in G taking into account one-way streets and the penalty p .

(a) (3 points) What are the nodes V_p ?

(b) (4 points) What are the edges E_p ?

(c) (3 points) What are the weights w_p ?

(d) (4 points) Given a node t , show how to reconstruct the shortest path with penalties from s to t in G using the solution to Dijkstra on G_p .

(e) (2 points) Identify a problem or algorithm we've seen in class (lecture, homework, or discussion) that influenced your work on this problem. Explain the attributes that are similar in the two problems and how that influenced your work. *Note: You may answer this question even if you did not solve the problem.*

PROBLEM 3 (Scheduling with preemption (16 points)). There are N jobs with running times $t_1, \dots, t_n$. Each job also has a **release time** of $r_1, r_2, \dots, r_n$, that is each job i cannot start until its release time r_i . Suppose also that we allow **preemption**, so that a task can be suspended and restarted at a later time. A schedule P is characterized by which job i is schedule at each unit time j . Let $c_i = \max_j(p_j = i)$ be the completion time of each job i .

- (a) (8 points) Give a **greedy** algorithm that produces the schedule P^* that minimizes the sum of the completion times $C = \sum_i c_i$.

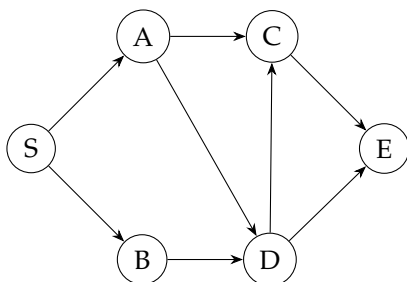
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(b) (2 points) State the running time of your algorithm.

(c) (6 points) Give a formal proof of correctness for your algorithm.

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PROBLEM 4 (Counting Paths (16 points)). You are given an unweighted, directed acyclic graph G and a starting node s . Your goal is to compute the number of paths from s to every other node in $O(|V| + |E|)$.



Node	S	A	B	C	D	E
# Paths	1	1	1	3	2	5

There are three paths from S to C

1. $S \rightarrow A \rightarrow C$
2. $S \rightarrow A \rightarrow D \rightarrow C$
3. $S \rightarrow B \rightarrow D \rightarrow C$

(a) (3 points) Define dynamic programming subproblems to solve this problem. Define each variable you introduce.

(b) (1 point) State any base cases.

(c) (4 points) Write a recurrence expressing the solution of the subproblem in terms of the solutions of smaller subproblems.

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- (d) (6 points) Write **pseudocode** for an $O(|V| + |E|)$ algorithm. Note: You must write a **dynamic programming** algorithm to receive credit. You can use algorithms from the course as black box.

- (e) (2 points) Identify a problem or algorithm we've seen in class (lecture, homework, or discussion) that influenced your work on this problem. Explain the attributes that are similar in the two problems and how that influenced your work. *Note: You may answer this question even if you did not solve the problem.*

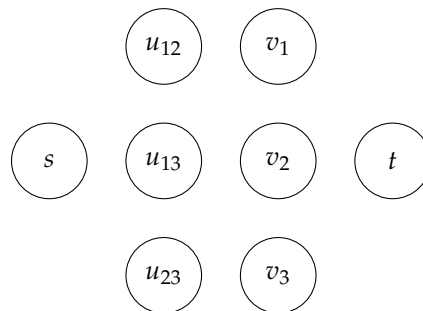
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PROBLEM 5 (Chances of Winning (16 points)). The CS softball team Runtime Terrors is not doing so well. Captain Matt West wants to know if there is a mathematical chance of winning the intramural league. There is a set of teams T other than Runtime Terrors. As input you are given a matrix $\text{games}[i][j]$ containing the number of games left between team i and team j and array $\text{score}[i]$ of the current score of each team. Runtime Terrors have another W games to play and a current score of P .

You need to design a **flow** algorithm to answer whether Runtime Terrors can at least tie for first place. You are given a black box algorithm P that takes in a graph $G = (V, E, c)$ and nodes s and t and returns the maximum $s - t$ flow.

The flow network that you will use will have the nodes $V = \{s, t\} \cup \{u_{ij} \mid \forall \{i, j\} \in \binom{T}{2}\} \cup \{v_i \mid \forall i \in T\}$.

Team Name	Score
Runtime Terrors	1
Bingo	3
Sabres	3
Costims	4



In this example, all pairs of teams have one more game to play. Notice that no matter the result of the games, one team will end up with 5 points, whereas Runtime Terrors can only get to 4.

(a) (4 points) Describe the edges between the nodes in V .

(b) (4 points) Give a capacity to each edge.

(c) (4 points) Give a condition between the network flow and Runtime Terrors' possibility of getting first place.

(d) (4 points) Given an outcome of the remaining games that places Runtime Terrors in first place, give an assignment to the network flow showing equivalence between the two problems.

PROBLEM 6 (NP-Completeness (16 points)). Traveling Salesman Problem (TSP) is an NP-Complete problem where you are given a complete, directed, weighted graph $G = (V, E, w)$ and asked to find the shortest simple cycle of length $|V|$. Consider TSP-1,2 which is a special case of TSP where all weights have value of one (1) or two (2).

(a) (4 points) Show that TSP-1,2 is in NP.

Hamiltonian Cycle is an NP-Complete problem where you are given a graph $G = (V, E)$ and must decide whether there exists a simple cycle over all vertices of G .

(b) (12 points) Give a polynomial time reduction from Hamiltonian Cycle to TSP-1,2.

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